Do NBA Player Statistics Predict Salary?

An Analysis of the 2023-24 NBA Season

MA346 - Project 2 | Spring 2026

Sago & Diego

Executive Summary

Our project takes a deeper look into whether in game performance statistics can predict NBA player salaries during and using the 2023–24 season. Using per-game statistics from Basketball Reference and salary data from Spotrac, we merged the two independent datasets and applied correlation analysis and multiple regression to answer our central question. The data consisted of covering 336 qualified NBA players from this season creating our central merged database.

Our key findings:

- Points per game appears to be the single strongest predictor of salary, with a correlation of r = 0.81.
- Minutes per game (r = 0.72) and assists per game (r = 0.60) are also strongly correlated with salary.
- A multiple regression model including and using seven performance statistics has an R^2 of 0.594 meaning these predictors explain approximately 59% of the variation in log-transformed salary
- Power forwards earn the highest average salary ($15.1M) out of any other position, while on the other hand shooting guards earn the least ($12.4M) on average.
- Several high-profile players such as Bradley Beal ($50.2M) and Zach LaVine ($44.5M) — appear significantly overpaid relative to their on-court production based on the average statistics.

These results show that performance does drive compensation in the NBA at least somewhat, but still 40% of salary variation is explained by factors outside of statistics. This includes variables such as market size, contract timing, positional scarcity, and star power.

1. Data Sources

1.1 NBA Per-Game Statistics — Basketball Reference

The first dataset was downloaded from Basketball Reference (basketball-reference.com), the gold-standard repository for NBA statistics. Basketball Reference has been the official NBA data partner since the 2018–19 season, sourcing its data directly from SportRadar, the NBA's official statistics provider. The dataset contains per-game averages for every player who appeared in the 2023–24 regular season, including points, rebounds, assists, steals, blocks, field goal percentage, and minutes played.

The dataset consists of 736 raw rows because players traded mid-season appear once per team plus a combined 'TOT' row. We kept only the TOT row for traded players, which ultimately left us with one record per player. After filtering for meaningful playing time (at least 10 games and 10 minutes per game), the cleaned stats dataset left us with 404 players.

1.2 NBA Player Salaries — Spotrac

The second dataset we obtained was from Spotrac (spotrac.com). Spotrac is one of the most widely cited sports contract databases in journalism and analytics and the website tracks only verified NBA contract information. We focused in on the data concerning each player's 2023–24 cap hit, contract type, and remaining years. It is used as a primary source by ESPN, The Athletic, and NBA front offices.

The salary dataset consisted of 450 players. After normalizing player names to match across datasets and performing an inner join, leaving us with 336 players appearing in both datasets with complete information. The 114 players not matched were either G-League or two-way contract players who do not have full stats entries in the Basketball Reference per-game table.

1.3 Why These Two Sources?

The project guidelines stated that we must collect data from two datasets that are different and independent of each other. With our sources, Basketball Reference (Sports Reference LLC) and Spotrac, these requirements are met as they are entirely separate companies with different data collection focuses and methods. We see that one is focused on game-level statistics while the other was centered around contracts and financial records. Since the player's name formats were different we see that we needed to clean the data sets for easier merge.

2. Data Cleaning & Preparation

Both datasets required cleaning before they could be merged. The steps we took are as described below.

2.1 Stats Dataset Cleaning

- Removed repeated header rows as the Basketball Reference inserted a header row every 25 rows in its CSV export.
- Converted all numeric columns such as points, rebounds, assists, etc from strings to floats using functions within pandas.
- Resolved the traded players issues keeping only the 'TOT' (combined) row for any player who appeared on multiple teams.

- Filtered to players with at least 10 games played and 10 minutes per game to exclude low-sample-size players.

2.2 Salary Dataset Cleaning

- Converted salary strings (e.g., '55,761,216') to integers by removing commas.
- Removed any duplicate entries that were found.

2.3 Merging the Datasets

Player names were normalized before merging to handle differences between the two sources:

- Removed accent characters (e.g., Luka Dončić became Luka Doncic).
- Standardized initials (e.g., R.J. Barrett became RJ Barrett).
- Removed name suffixes (Jr., Sr., II, III) which were inconsistently applied across sources.
- Applied common nickname fixes (e.g., Nicolas → Nic, Herbert → Herb).

After this cleaning we did an inner join to keep only players present in both datasets, leaving us with the final merged dataset. The finalized dataset contains 336 players from all 30 NBA teams.

Metric	Stats Dataset	Salary Dataset	Merged Dataset
Raw rows	736	450	—
After cleaning	404	450	—
After merge (inner join)	—	—	336
NBA teams represented	30	30	30
Columns	15	4	16

3. Analysis & Results

3.1 Research Question

Do NBA player performance statistics predict their salary? If so, which statistics are most predictive, and how well do statistics collectively explain a player's salary variation?

3.2 Salary Distribution

Before analyzing relationships, we examined the distribution of salaries. The raw distribution is heavily right-skewed, which basically means that most players earn relatively modest salaries while a small number of stars earn extremely high contracts. This is expected as there are more role players in the league as there are superstars. However, for the statistical tests we will be using, it's important that the data is approximately normally distributed, not skewed. Therefore, we can apply a log transformation in order to make the data more normally distributed as seen below.

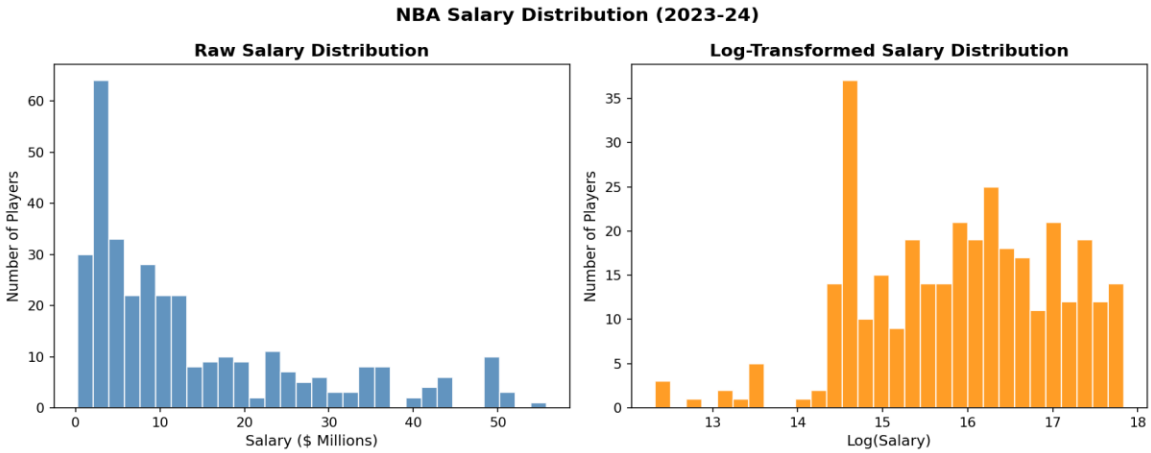

Figure 1: The log transformed data on the right is more normal than the original data on the left

The average salary among qualified players was $13.7M, with a median of $8.9M. The highest earner was Stephen Curry at $55.8M, while the minimum salary in the dataset was $223,718 for Jordan Goodwin.

3.3 Correlation Analysis

We computed Pearson correlation coefficients between each per-game statistic and raw salary. All correlations reported below are statistically significant ($p < 0.001$).

Statistic	Correlation with Salary (r)	Interpretation
Points Per Game	0.81	Very Strong
Minutes Per Game	0.72	Strong
Assists Per Game	0.60	Strong
Rebounds Per Game	0.45	Moderate
Games Played	0.22	Weak
Steals Per Game	0.38	Moderate
Blocks Per Game	0.27	Weak-Moderate

Points per game is by far the strongest individual predictor of salary, which checks out. Scoring is the most visible and marketable skill, and teams pay a premium for reliable scorers. At the same time, the players who score the most points per game are often times seen as the best players, which correlates with them having the highest salaries. Minutes played is nearly as correlated, which reflects the fact that coaches give the most minutes to the players they value most, and therefore, pay the most.

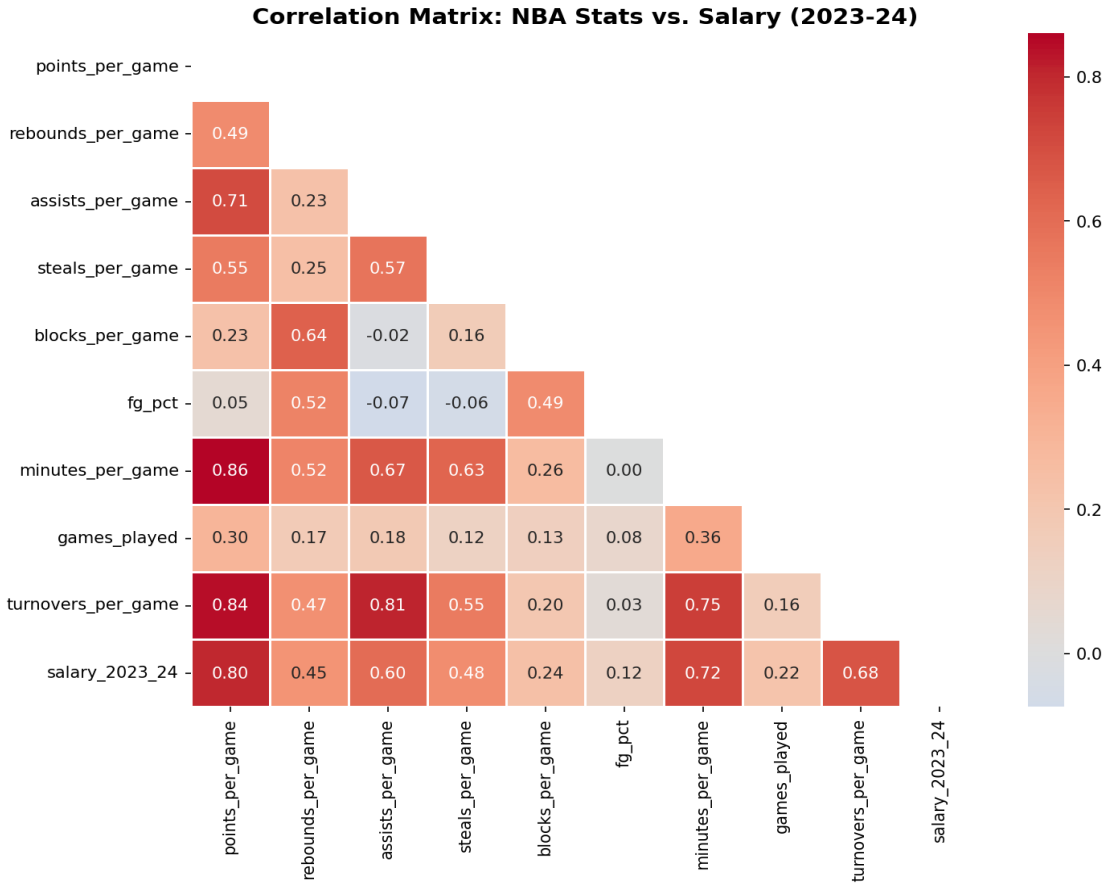

Figure 2: Correlation matrix of all performance statistics and salary. Darker red indicates stronger positive correlation.

3.4 Points Per Game vs. Salary

The scatter plot below shows the relationship between points per game and salary, colored by position. The positive linear trend is clear across all positions, though there is substantial variation around the trend line. This confirms that scoring alone does not fully determine salary, but it does have correlation.

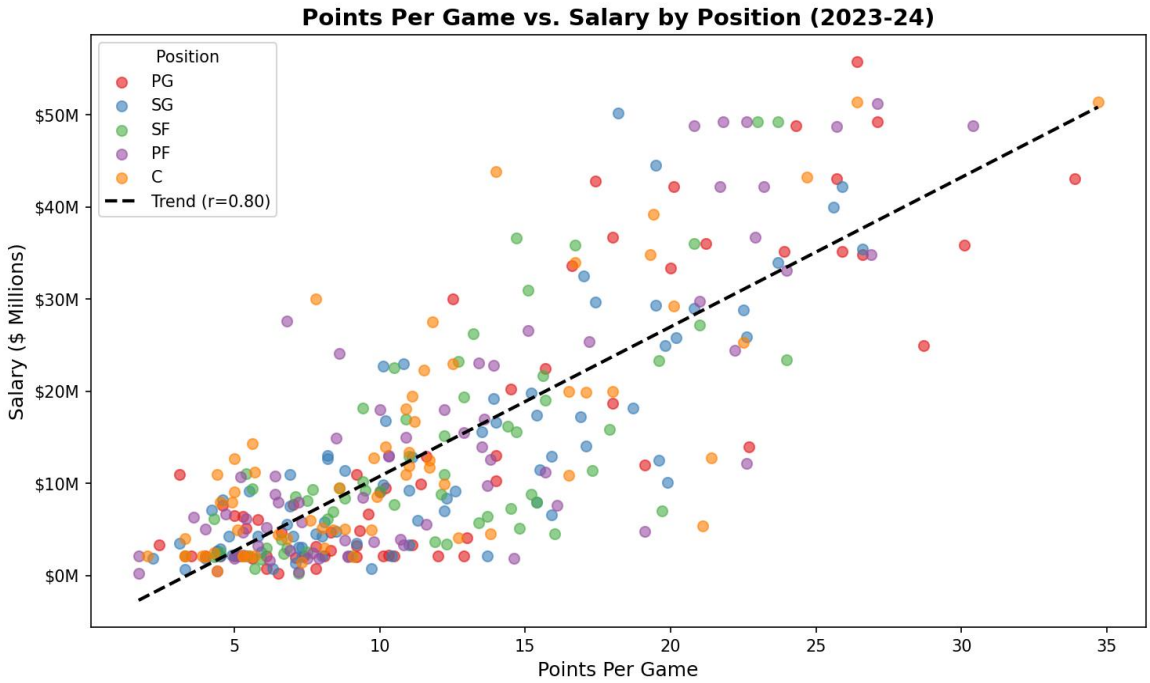

Figure 3: Points per game vs. salary by position.

3.5 Salary by Position

Power forwards and point guards, with averages of $15.1 and $14.8 million respectively, command the highest salaries on average, while shooting guards earn the least at $12.4 million. The gap between mean and median salaries is largest for point guards, reflecting a few superstar contracts, such as Curry, Lillard, and Doncic, pulling the mean upward.

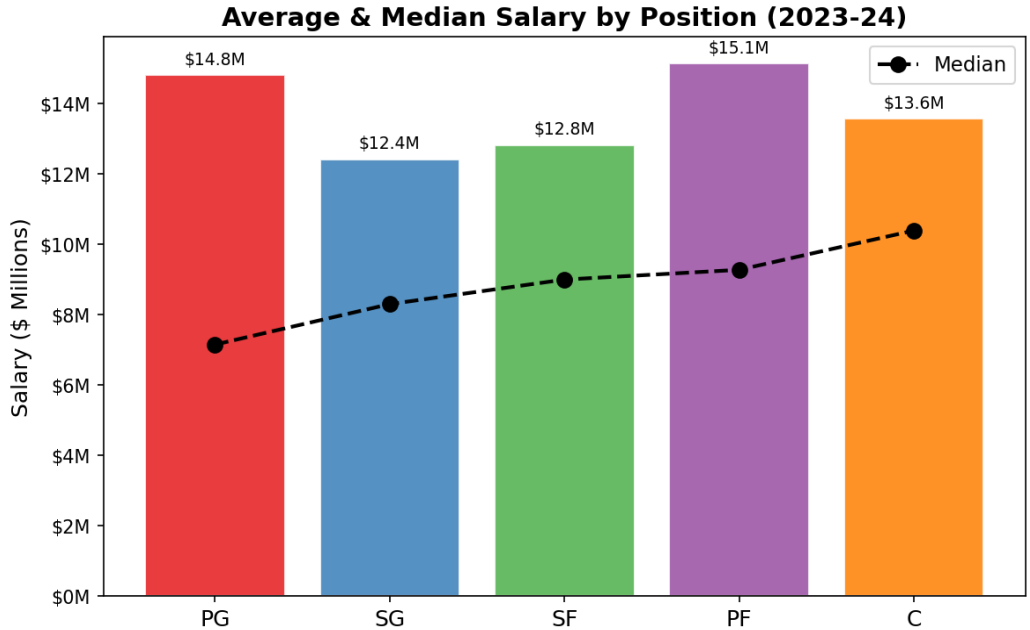

Figure 4: Average and median salary by position. Dots show median while bars show mean.

3.6 Multiple Regression

To assess how well statistics collectively explain salary, we fit a multiple linear regression with the new log salary as the outcome and seven performance statistics as predictors: points, rebounds, assists, steals, blocks, minutes per game, and field goal percentage.

The model achieved an R^2 of 0.594, meaning that these seven performance statistics together explain approximately 59% of the variation in salary. This is a meaningful result since it confirms that on court production is the primary driver of NBA compensation. However, the remaining 41% of variation reflects factors outside the statistics box score. Some of these variables may include the market size or even injury history as teams may shy away from commiting a lot of money to players who have a history of being injured.

3.7 Player Value Map

To visualize which players deliver the best and worst value relative to their salary, we created a composite performance score, 40% points + 20% rebounds + 20% assists + 10% steals + 10% blocks, and then we plotted it against salary. Dashed lines divide players into four quadrants at the median salary and median performance score.

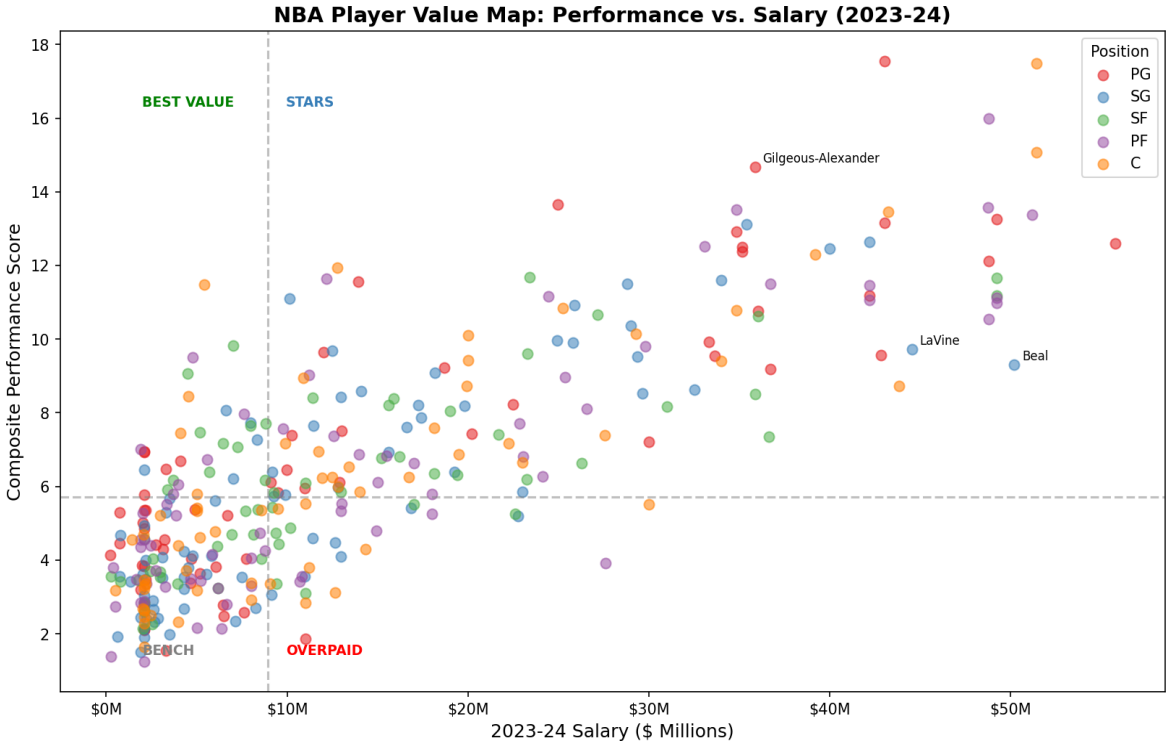

Figure 5: Player value map. Upper-left = best value. lower right = overpaid relative to performance.

Nikola Jokic stands out as arguably the league's best value as he ranks among the top performers while earning $51.4 million which sounfs like a high salary but is justified by his historic production.

Bradley Beal, $50.2 million, and Zach LaVine, $44.5 million, appear in the lower-right quadrant, earning max-level salaries with below-average performance scores, making them outliers in terms of value.

3.8 The Top-Paid Players

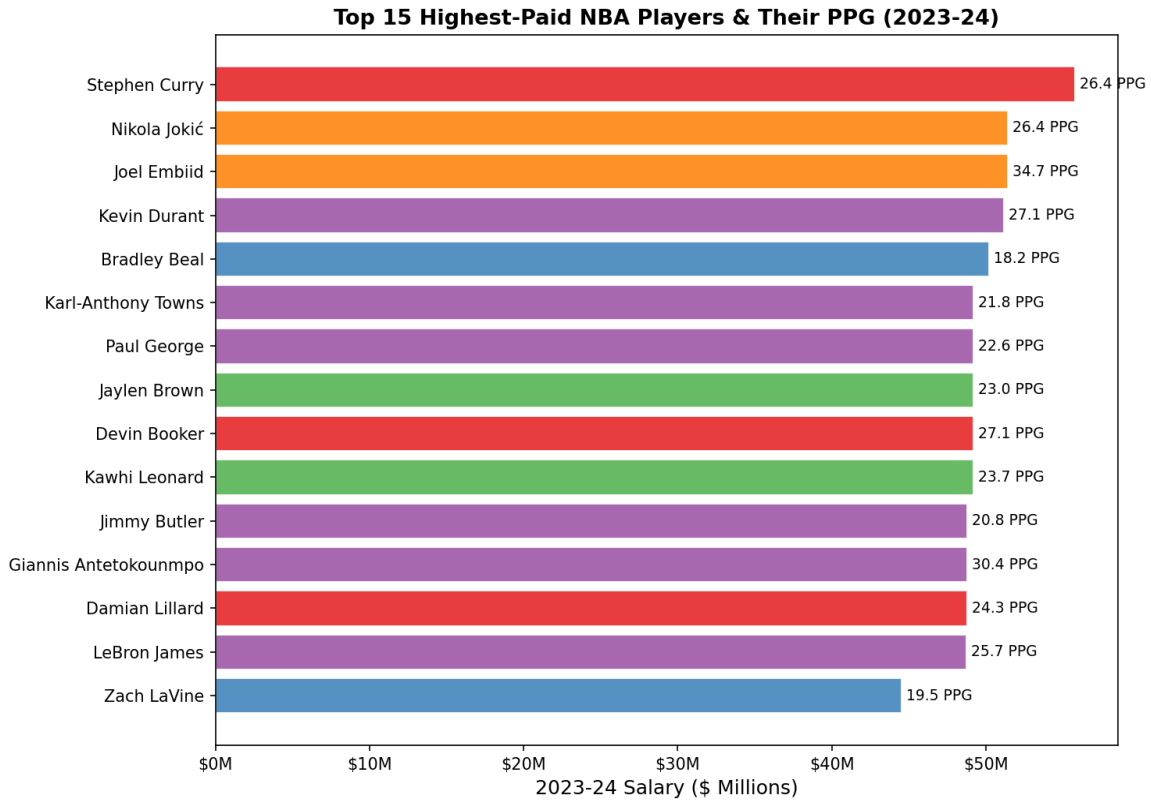

Figure 6: The 15 highest-paid NBA players and their points per game. Bradley Beal's 18.2 PPG stands out as low relative to his $50.2M contract.

4. Conclusions

This project set out to answer one question, do NBA player statistics predict salary? The answer is yes, substantially but not completely.

Points per game is the single most powerful predictor of salary, followed by minutes per game and assists. A model using seven performance statistics explains 59% of salary variation, which is strong for a real-world dataset where human judgment, contract history, and market forces all play a role.

The most practically interesting finding is the value gap. There are some players earning max contracts, such as Bradley Beal and Zach LaVine, who produced at levels far below what their salary would predict, while players on minimum deals, like Jordan Goodwin and Lamar Stevens, provided exceptional production per dollar. This has real implications for NBA front offices since teams that can identify undervalued players gain a competitive advantage.

For future work, this analysis could be extended in several ways. One way would be by incorporating some more advanced metrics, such as player efficiency rating or VORP. These metrics may explain more salary variance than the more simple per game stats since these are the ones organizations are looking deeper into. Additionally, adding contract year could be important. There may be players who are playing on contracts which they signed five years ago, and are no longer playing at the same level as they were when they signed. Lastly, an interesting thing to look at as well is seeing whether over and under paying players correlate with teams win – loss records.

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5. Data Sources & References

- Basketball Reference (2024). NBA Per-Game Statistics 2023–24 Season. Sports Reference LLC. https://www.basketball-reference.com/leagues/NBA_2024_per_game.html
- Spotrac (2024). 2024 NBA Cap Hit Rankings. https://www.spotrac.com/nba/rankings/player/_/year/2024/sort/cap_total
- Sports Reference LLC (2018). Basketball Reference Now Using Official NBA Data. https://www.sports-reference.com/blog/2018/12/basketball-reference-now-using-official-nba-data/